100X Improvement in Bioelectronic Signal Acquisition

Device and scaling findings: IGZO TFT extraction and N-channel scaling

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Generated 2026-08-19 - ngspice-46, closed-loop verified

Headline

The topology is sound and reproduces its verified design point. Two things block it: the shared return destroys the channel plan, and the measured IGZO TFTs fall 9x-64x short of the required transconductance at 1.5 V.

1. Isolated return is a requirement, not an optimisation

With the shared return the resonant modes land 27-38 % away from their design frequencies, and the error grows with channel count. With an isolated return the error is <= 2.4 % and shrinks with N (1.2 % at N=10).

2. The IGZO TFTs are normal devices; the circuit asked for silicon

Measured Kp = 0.72 uA/V^2 (best chip, long channel) against 1000 uA/V^2 assumed in every simulation to date - a factor of ~1400. This is not a fabrication defect: it is what mu*Cox gives for a normal IGZO TFT.

3. The wall is the supply voltage, not the current budget

At VDD = 1.5 V the transistor cannot pass the needed current at all:
gm saturates at 89 uS (best case) against 808-2019 uS required.
More current does not help - VGS is capped by the supply.

4. Contacts, not the semiconductor, limit the short-channel devices

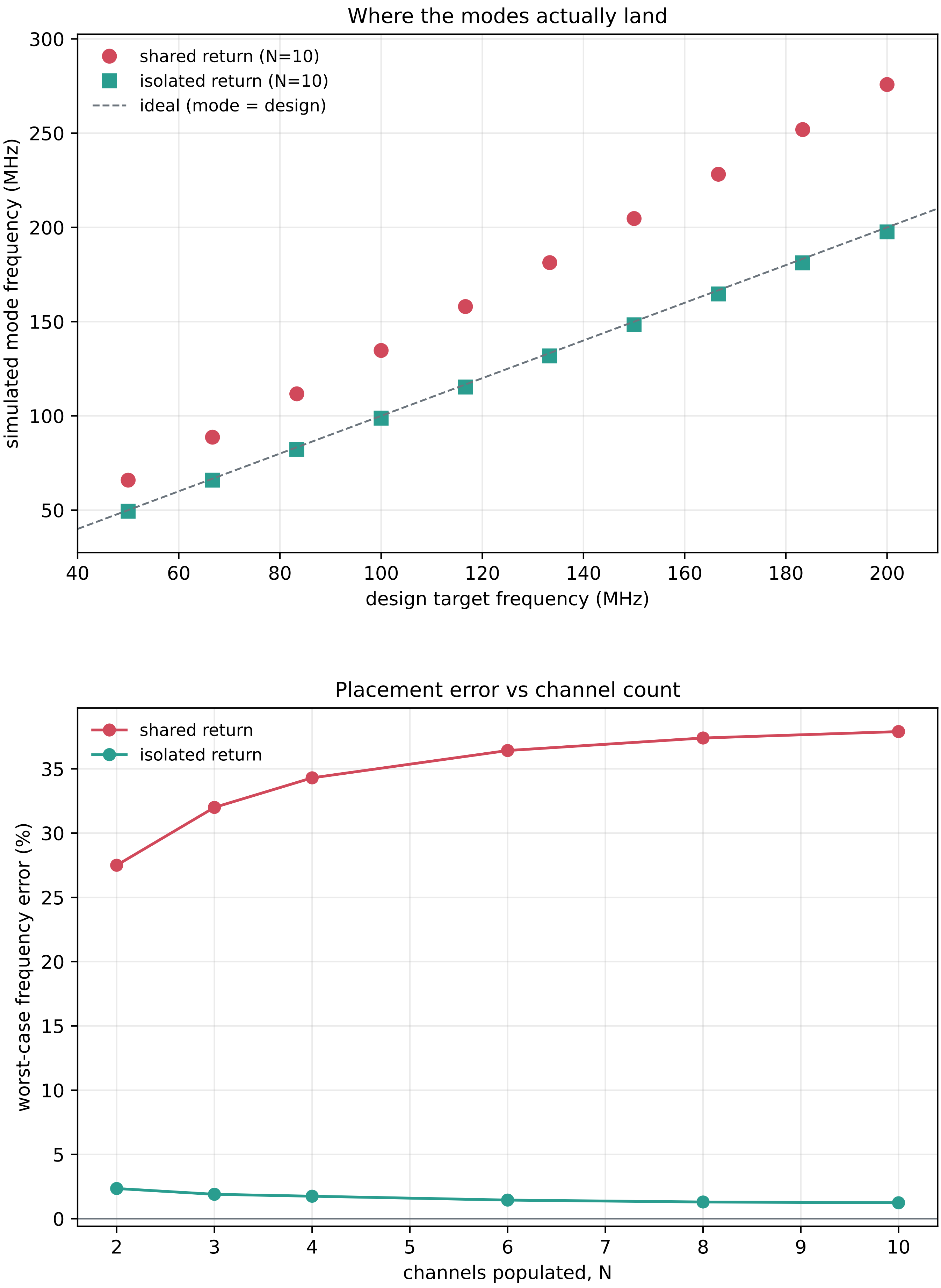
Kp is flat at ~0.72 uA/V^2 for L >= 40 um and collapses to 0.13 uA/V^2 at L = 8 um. Linear and saturation extractions agree to 0.5 % on long channels and diverge to 0.64 on short ones - the contact fingerprint.
2*Rc*W = 6.6e6 Ohm*um (660 Ohm*cm), ~100x worse than good IGZO practice.

5. gm required scales with tank loss, and that lever is too small

Measured gm_req / RS = 854 uS/Ohm, constant to +-4 % over a 16x range of RS. Closing a 9x gap this way needs RS = 0.11 Ohm; thin-film inductors go the other way.

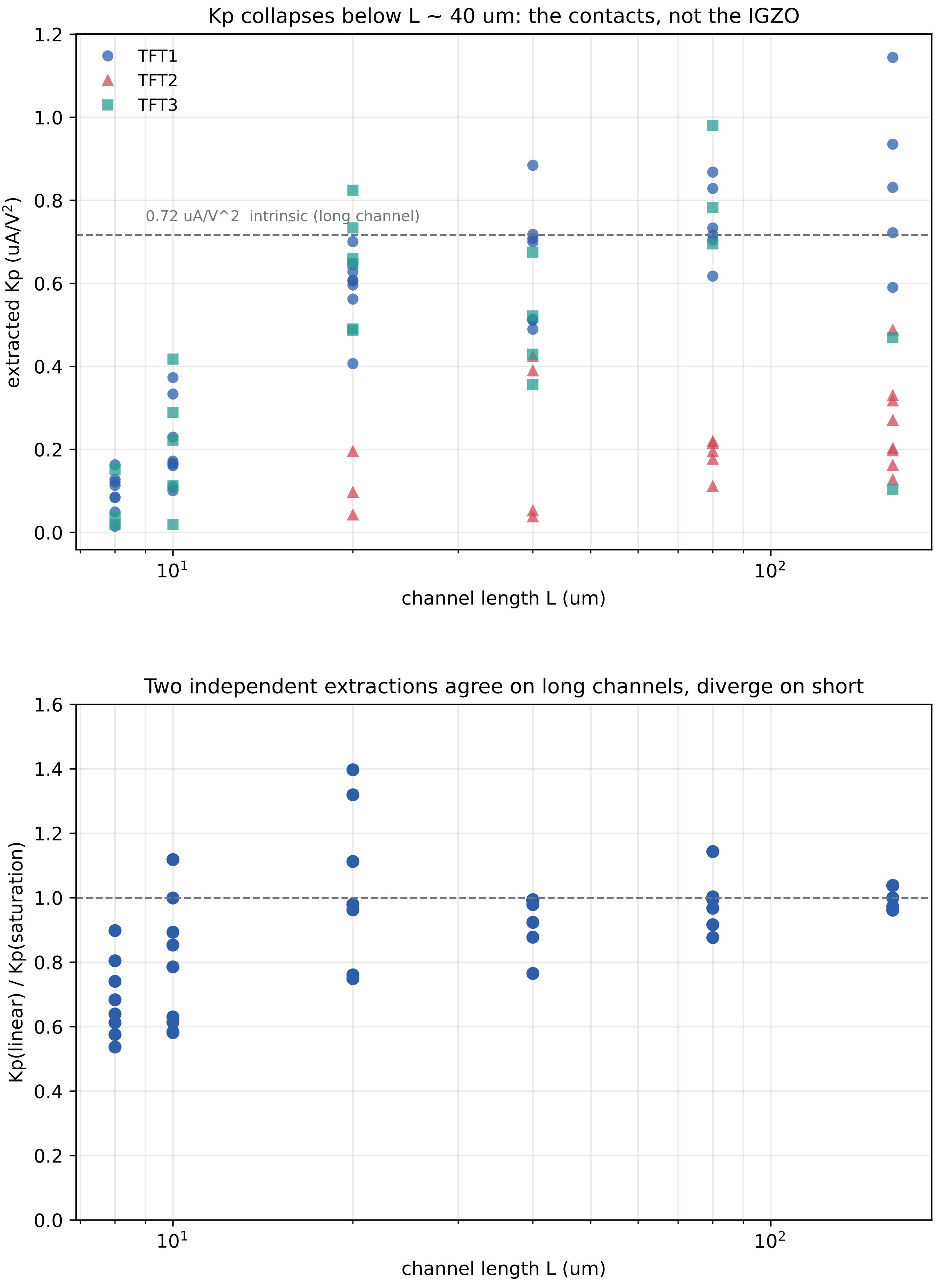
Every number here was measured in this session: the .xls readers were cross-checked against the pre-existing reports (172/172 exact) and the netlist generator reproduces the verified two-channel design point.

Phase 1 / 2 - Channel scaling: shared vs isolated return



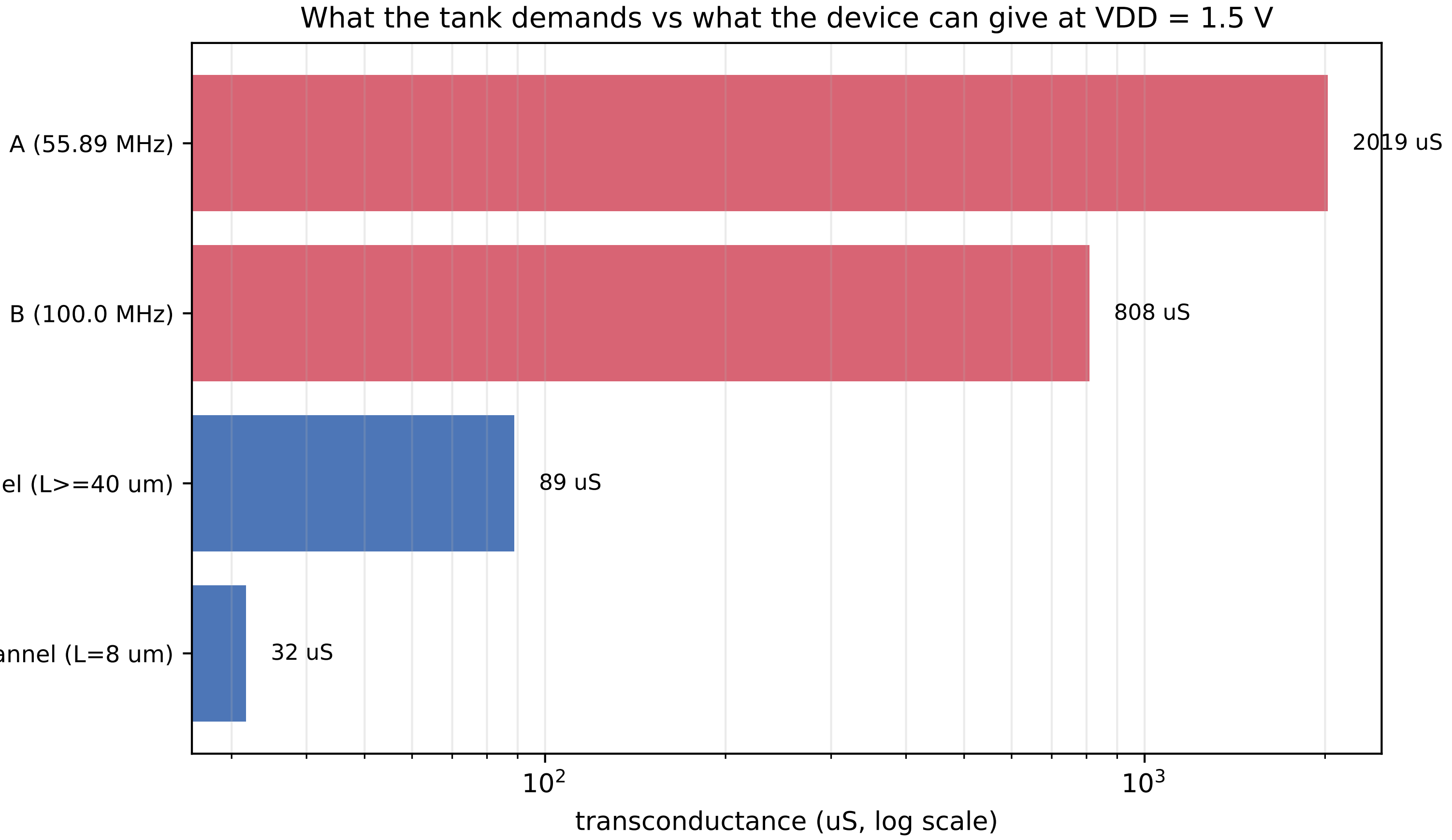
Shared return: the N return inductors sit in parallel, so the effective return inductance is (L/2)/N and shrinks as channels are added. The 16.67 MHz design spacing is replaced by a ~23 MHz spacing the designer does not control, and channel 10 lands at 275.8 MHz - outside the 50-200 MHz band entirely.

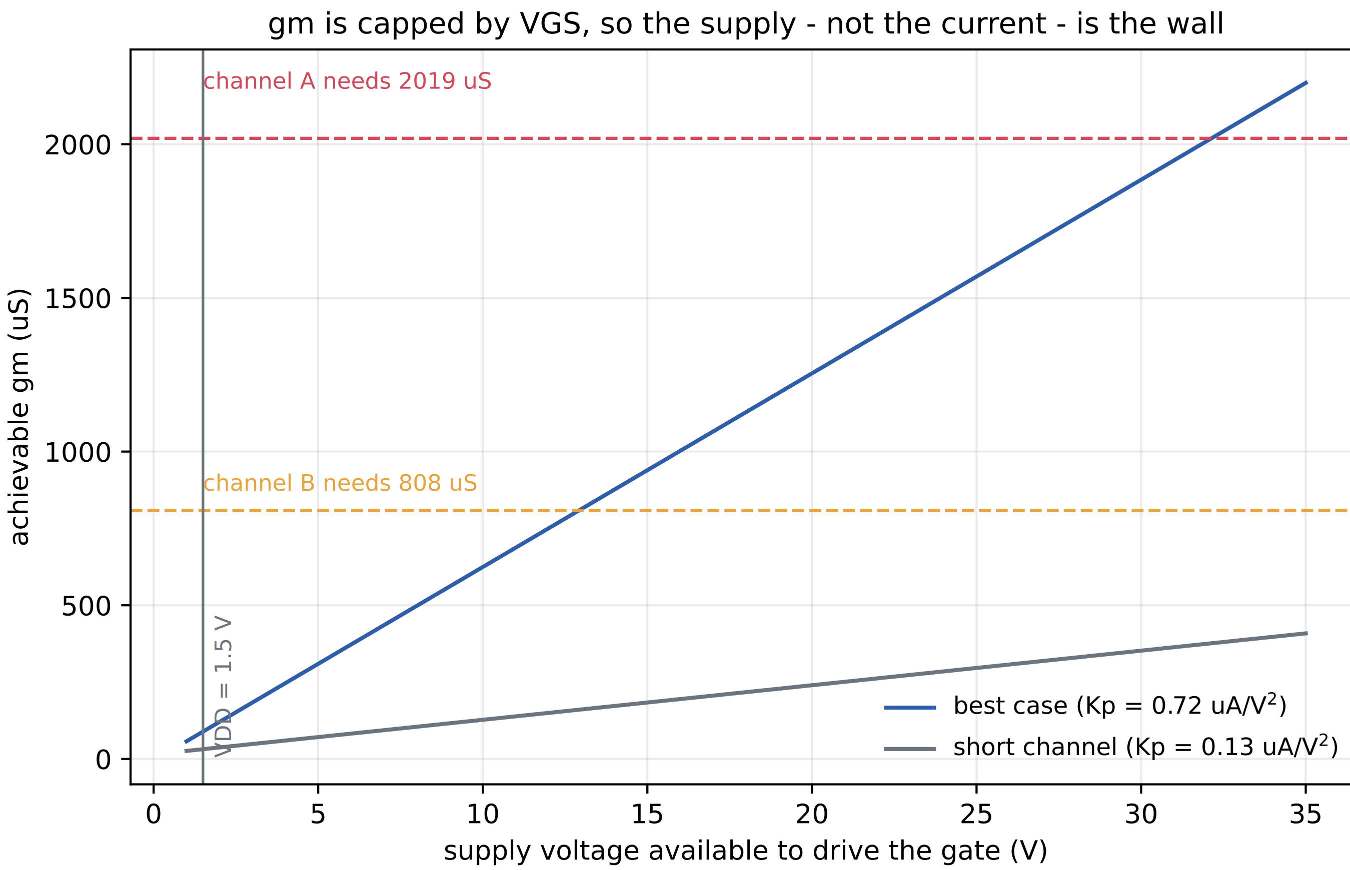
Phase 5 - IGZO TFT extraction: contact-limited, not material-limited



The linear route (gd = dId/dVDS as VDS -> 0) and the saturation route (sqrt(Isat) vs VGS at VDS = 10 V) agree to 0.5 % for L >= 40 um. Below that the linear route reads low, which is exactly how series contact resistance shows up. Independently: 2*Rc*W = 6.6e6 Ohm*um, contact-limited length L_c ~ 54 um.

Phase 5 - The transconductance gap





W = 703 um, L = 8 um: the largest pair that fits half of a 150 x 150 um cell.
Channel B would close at VDD ~ 13 V, channel A at ~ 32 V - both far above the
1.5 V rail, and the resulting bias current then exceeds the 100 uW/channel
budget by three orders of magnitude.

What to do next

Immediate - costs nothing, changes everything

Move to the isolated return before any further channel work. Every downstream phase (co-tuning, AGC, power) gets simpler because the channels stop moving each other's resonances.

Decide the supply voltage question

IGZO TFTs are normally run at 5-20 V, not 1.5 V. The 1.5 V rail is inherited from a silicon-CMOS assumption. Either raise the rail (and re-do the power budget) or accept that this topology needs a different active device. This is a project-level decision, not a design tweak.

Fix the contacts before fabricating more devices

2*Rc*W = 660 Ohm*cm is ~100x off good IGZO practice. Recovering that alone lifts short-channel Kp from 0.13 to ~0.72 uA/V^2 - a 5.6x gain in Kp, i.e. 2.4x in gm, for no area and no power.

Measure what is missing

The VGS/, COX/ and HALL/ folders are empty for all three chips, and the CAP structure area is not recorded. Without Cox we cannot separate mu from Cox, and - more urgently - we cannot predict the gate capacitance that loads the tank at 50-200 MHz. Also record W and L for the devices in TFT/test/: their transfer sweeps are the best data in the set and currently cannot be normalised.

Open question this session could not settle

Channel-to-channel interaction could not be measured: AC analysis is linear and cannot see oscillation, so the bias threshold sweep never triggers. This needs pole-zero (.pz) or transient analysis. The 9 % interaction figure in CLAUDE.md should be re-derived that way.

Verification performed

BIFF8 .xls reader vs pre-existing Isat_report.xlsx: 172/172 exact.
Transfer extraction self-consistency at VDS = 0.1 / 0.1 / 1 / 2 V:
    26.4 / 27.0 / 27.1 / 28.4 uA/V^2 (7 % spread over 20x in VDS).
Extracted Vth reproduces the instrument's own VT column.
Netlist generator reproduces the verified two-channel design point:
    A 55.8901 MHz / 35.90 dB / 22.98 kHz / Q_L 2432
    B 100.0021 MHz / 39.69 dB / 13.98 kHz / Q_L 7151
    (documented: 55.890 / 35.9 / 23.2 / 2413 and 100.002 / 39.7 / 13.9 / 7217)